

9 **(long)**

(a) What is the difference between asynchronous and synchronous sequential circuits, and give one advantage and disadvantage of each. [2]

(b) A modulo-6 synchronous binary counter is to be designed using only J-K bistables and 2-input NOR gates. If an unused state occurs, the counter returns to the state 0 on the next clock pulse.

(i) Draw a state table for the counter. [5]

(ii) Find the required J-K logic inputs for each bistable. [5]

(iii) Draw the complete circuit. [5]

(iv) If other 2-input gates were also available, show how the design could be changed to use a smaller total number of gates, and list the gates used. [3]

(c) Implement a J-K bistable with a D bistable and a minimum number of AND, OR and inverter gates. Draw the circuit to show your implementation. [10]